COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	July 17, 2026	**DATE SUBMITTED:**	July 17, 2026
NAME:	Quiming, Kadaphy James A>	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. HOA 2.1 Arrays, Pointers, and Dynamic Memory Allocation

1. PROCEDURE OUTPUT

Screenshot

```cpp
#include <iostream>
#include <string.h>

class Student{
    private:
        std::string studentName;
        int studentAge;

    public:
    // Constructor
    Student(std::string newName = "John Doe", int newAge = 18) {
        studentName = std::move(newName);
        studentAge = newAge;
        std::cout << "Constructor was called.\n";
    }

    // Deconstructor
    ~Student(){
        std::cout << "Destructor was called.\n";
    }

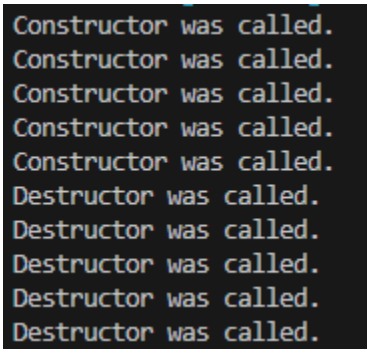
    // Copy Constructor
    Student(const Student &copyStudent){
        std::cout << "Copy Constructor  was called.\n";
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    // Display Attribute
    void printDetails(){
        std::cout << this->studentName << " " << this->studentAge << std::endl;
    }
};
```

```
Constructor was called.
Copy Constructor  was called.
Constructor was called.
Destructor was called.
Destructor was called.
Destructor was called.
```

Observation	I observed that the destructor was called 3 times because the constructor was called twice and copy constructor once.

Table 2-1. Initial Driver Program

Screenshot	 
Observation	The constructor was called 5 times because there are 5 objects in the array , then the destructor was also called 5 times.

```
int main() {

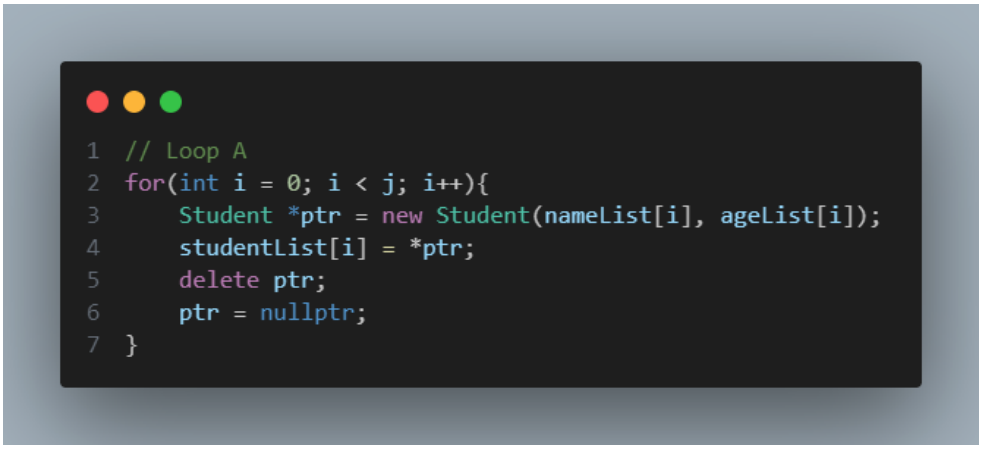
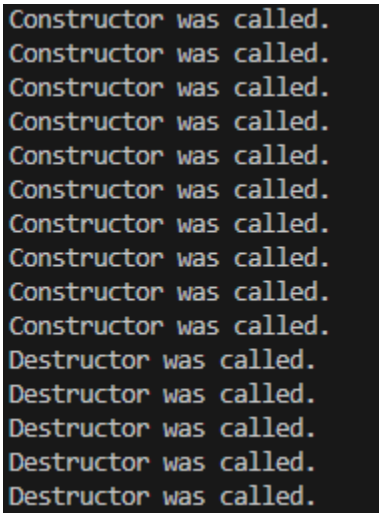
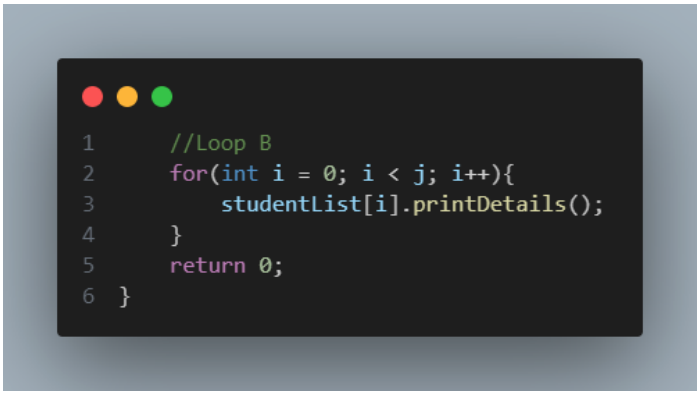
    const size_t j = 5;

    Student studentList[j] = {};
    std::string namesList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};

    return 0;
}
```

```
Constructor was called.
Constructor was called.
Constructor was called.
Constructor was called.
Constructor was called.
Destructor was called.
Destructor was called.
Destructor was called.
Destructor was called.
Destructor was called.
```

Table 2-2. Modified Driver Program with Student Lists

Loop A	 
Observation	For Loop A 5 strings was created in nameList and 5 integers was created in ageList, then they were combined like (“Name”, Age) so the Destructor was only called 5 times instead of 10. I also added delete ptr; and ptr = nullptr; 😀
Loop B	

	Carly 15 Freddy 16 Sam 18 Zack 19 Cody 16
Observation	Loop B was just there for the printing of the names and age
Output	

```cpp
int main() {

    const size_t j = 5;

    Student studentList[j] = {};
    std::string nameList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};

    // Loop A
    for(int i = 0; i < j; i++){
        Student *ptr = new Student(nameList[i], ageList[i]);
        studentList[i] = *ptr;
        delete ptr;
        ptr = nullptr;
    }

    //Loop B
    for(int i = 0; i < j; i++){
        studentList[i].printDetails();
    }
    return 0;
}
```

```
Constructor was called.
Constructor was called.
Constructor was called.
Constructor was called.
Constructor was called.
Destructor was called.
Constructor was called.
Destructor was called.
Constructor was called.
Destructor was called.
Constructor was called.
Destructor was called.
Constructor was called.
Destructor was called.
Carly 15
Freddy 16
Sam 18
Zack 19
Cody 16
Destructor was called.
Destructor was called.
Destructor was called.
Destructor was called.
Destructor was called.
```

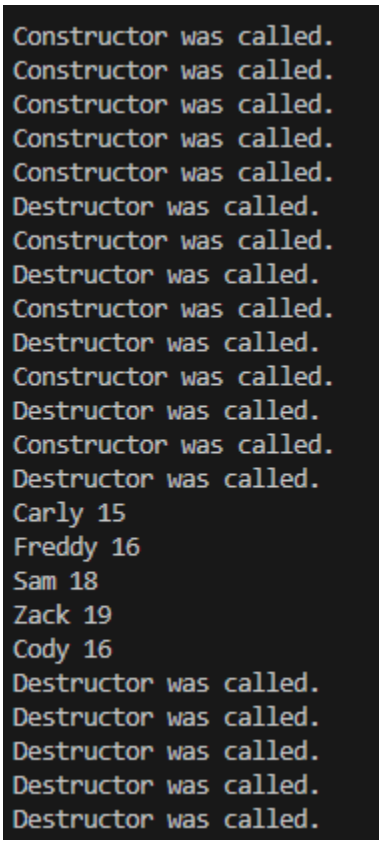
	
Observation	The first 5 constructor was being called because studentList is equal to 5, but it's only creating 5 (“John Does”, 18), then the alternating of constructors and deconstructors creates the listed names and age, putting them in a heap then deleting them after printing. After all that everything that is in studentList gets destroyed hence the 5 Destructor was called message.

Table 2-3. Final Driver Program

Modifications	
Observations	Added these two lines in loop A for optimizing.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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Purpose of Use: **Code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I only entered the error codes, not the code itself. I also asked it to not give me codes and just explain what the problems are.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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